A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 14

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1 Primary:** succession on a newly created habitat with no life / no soil (bare rock, new pond, sand) **1 Secondary:** succession restarting after an existing community is disturbed (tree felling, ploughing, fire, end of grazing) **1** Faster because soil already exists and many seeds / roots / organisms are still present so there are fewer delays for colonisation **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 3 points below. **1** Human activity (grazing, mowing, burning, coppicing) regularly removes the later successional species so the climatic climax cannot form **1** The stable community that develops under this repeated disturbance is a plagioclimax **1 Examples:** chalk grassland / hay meadows maintained by grazing or mowing – lowland heath by burning – coppiced woodland – Machair grassland – Broads grazing marshes **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **deflected succession** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3
01.3	**1 mark each** 1 mark for each of the 2 points below. **1** If the traditional activity stops, secondary succession returns the habitat to scrub / woodland and the plagioclimax species are lost **1** Rainforest and coral reefs are climax communities so conservation aims to minimise human impact **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
01.4	**1 mark each** 1 mark for each of the 2 points below. **1** Extreme environments (deserts, polar) are dominated by abiotic factors – few species tolerate them – populations fluctuate and food webs are simple so changes in one food species have large effects **1** Stable environments allow many species to evolve inter-species relationships – populations are regulated by biotic factors – complex food webs give stability (rainforests, reefs) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 76-77; AQA spec 3.1.3.3.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**2 marks per point** 2 marks for each of the 3 points below; award 1 mark where a point is only partly made. **2** Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive **2** Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light – diversity falls **2** Hedgerow removal destroys habitat / corridors and nesting sites – loss of natural pest predator habitats increases pest damage – larger fields increase wind erosion – genetic contamination and introduced species also accepted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	6
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows **1** These communities depend on the farming continuing **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity	2
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** CO2 from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery – ploughing increases soil respiration **1** Methane from livestock intestines and rice paddies – nitrous oxide from manure and nitrogen fertilisers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **embodied energy** → all the energy used to make and transport a product	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	
03.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Environmental DNA (eDNA) analysis	1
03.2	**1 mark each** 1 mark for each of the 2 points below. **1** Transmitters are too heavy for small birds / their batteries would not last for a long-term study **1** Geolocator tags are small and only record time and light levels – latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Each separate (regional) population has its own unique gene pool **1** DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
04.1	**1 mark each** 1 mark for each of the 4 points below. **1** Trees (hazel, oak) are cut to ground level every 3-30 years and allowed to regrow from the stump **1** About 0.5 ha cut each year on a rotation (eg 8 years) so the wood is a patchwork of areas of different ages **1** Each age has its own community: newly cut areas get high light for wildflowers and butterflies – older areas give dense cover for nesting birds / dormice **1** Pollarding cuts above browsing height so deer / livestock cannot eat the regrowth **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	4
04.2	**1 mark each** 1 mark for each of the 4 points below. **1** Wolves (top predator, exterminated by 1926) reduced deer populations and changed their behaviour **1** Fewer young trees eaten so trees and riverside vegetation recovered **1** More trees allowed beaver populations to increase – beaver dams increased wetland area **1** Wetland species (birds, fish, invertebrates) became more common – overall biodiversity rose, though wolves cause problems for livestock farmers **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity	4

Do not write outside the box

Question	Marking guidance	Marks
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** Undesirable species (introduced predators / competitors, or natural succession species such as scrub, conifers in the New Forest heathland) reduce survival of the target species **1** Removal restores suitable conditions (eg heathland for sand lizards and smooth snakes) or reduces predation / competition **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3-3.1.3.4.	
05.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B **Also accept** • **abiotic** → non-living / physical factors	1
05.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Deforestation **1** Ploughing of grassland / expansion of farmland **1** Reservoir creation / flooding **1** Mineral extraction, especially open-cast mining **1** Urban expansion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. • 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 20-26; AQA spec 3.1.2.2.	

Do not write outside the box

Question	Marking guidance	Marks
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish **1** Species occupy different niches / trophic levels so they do not compete and all food sources are used **1** Bottom feeders re-suspend nutrients boosting plankton / plant growth **1** Predators control over-breeding species (tilapia) so remaining fish grow larger – total yield exceeds any single species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
06.2	**1 mark each** 1 mark for each of the 4 points below. **1 IMTA:** fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) **1** Wastes of one species become nutrients for another so less pollution and extra harvests **1 Aquaponics:** aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) **1** Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish) – rice-fish paddies similar **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages **1** Production still depends on (and removes juveniles from) the wild population – carnivores need large fish inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2

Do not write outside the box

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	
07.1	**1 mark each** 1 mark for each of the 3 points below. **1** Chicken and salmon need far less feed per kg gained (1.2-1.8 vs 7.0) **1** Fish are ectotherms so no energy is used for thermoregulation and water supports their mass – poultry grow fast and are kept warm indoors with movement restricted **1** Cattle are large endotherms, use energy in movement, and lose energy in methane from cellulose digestion **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
07.2	**1 mark each** 1 mark for each of the 3 points below. **1** Control of movement (housing / restricted grazing) so less energy is used in muscle activity **1** Temperature control (heated / insulated buildings) so less energy is used in maintaining body temperature **1 Species selection:** choose species with low FCRs (poultry, fish) – high-energy easily digested diets – slaughter at the fastest growth stage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
07.3	**1 mark each** 1 mark for each of the 4 points below. **1** Only about 10% of the energy at one trophic level is passed to the next – most is lost as heat in respiration / metabolism or in undigested material **1** Humans eating plants (primary consumers) receive more of the energy captured by photosynthesis than eating animals fed on those plants (secondary consumers) **1** Livestock better where land cannot grow crops (upland, semi-arid, steep, thin soils): grazing animals with symbiotic gut bacteria digest cellulose in grass that humans cannot **1** Omnivores (pigs) convert food wastes (dairy by-products, catering waste) into meat **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **trophic level** → the feeding level in a food chain • **symbiotic** → a close relationship in which both species benefit **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 326-328; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 5 points below. **1 Timber:** building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood) **1 Fuel:** fuelwood and charcoal - still the main energy source for most people in LEDCs **1 Food:** fruit, nuts, bushmeat, livestock fodder – crops of forest origin (coffee, cacao, bananas, Brazil nuts) – pigs and chickens were forest animals **1 Fibres:** paper (wood pulp cellulose), cotton, viscose / rayon **1 Medicines:** quinine (cinchona), taxol (yew), aspirin (willow) – most species not yet researched **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	5
08.2	**1 mark each** 1 mark for each of the 3 points below. **1** Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time – total forest carbon (vegetation plus soil) is about double that in the atmosphere **1** Growing trees sequester CO_2 by photosynthesis and release oxygen – forest soils hold more carbon than the trees (especially boreal) **1** Planting / protecting forests removes CO_2 and reduces the enhanced greenhouse effect **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **residence time** → the time a molecule stays in the atmosphere / atmospheric lifetime • **enhanced greenhouse effect** → extra warming caused by added greenhouse gases **Do not credit** • 'the greenhouse effect is bad for the planet' - the natural greenhouse effect keeps the Earth habitable; it is the enhanced effect that causes the problem • a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation	3

Do not write outside the box

Question	Marking guidance	Marks
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** Highest transpiration of any ecosystem increases downwind rainfall – interception increases evaporation – deep organic soils retain water and delay runoff so river flow is more even and flooding reduced **1** Low albedo absorbs sunlight – heat stored and released at night reduces day-night temperature extremes – canopy microclimate: lower light, wind, higher humidity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1.	
09.1	**1 mark each** 1 mark for each of the 4 points below. **1** k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses **1** Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population **1** Deep-water aggregations easily located and caught by trawling **1** Recovery takes decades so fisheries have had to be closed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	4
09.2	**1 mark each** 1 mark for each of the 2 points below. **1** Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young **1** Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2

Do not write outside the box

Question	Marking guidance	Marks
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort **1** Masks the decline of the breeding population until a sudden collapse **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Neither is a target species but both are caught in demersal trawls – skate now extinct in the Baltic and critically endangered elsewhere **1** Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	
10.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Zooxanthellae are photosynthetic algae living inside the polyp cells **1** The algae absorb CO_2 and excretory wastes from the polyp for photosynthesis and growth (efficient nutrient recycling) **1** They supply the polyp with oxygen and glucose - over 95% of its food energy **1** In return the algae gain a protected habitat and nutrients they could not absorb from the nutrient-deficient water **1** Polyps also feed on plankton captured by cilia and nematocysts **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **recycling** → reprocessing a material to make something new **Do not credit** • 'recycling uses no energy' - collection, sorting and reprocessing all use energy, usually less than making the material from new	4
10.2	**1 mark each** 1 mark for each of the 2 points below. **1** Corals deposit a skeleton of calcium carbonate (calcite) **1** Each carbonate molecule requires the absorption of a molecule of carbon dioxide, so reefs are a carbon sink **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
10.3	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Importance:** cover 1% of ocean area but support over a quarter of marine species – fisheries (fish, lobsters, conch) – medicines (AZT, Ara-C, dolastatin, secosteroids, bryostatin, trabectedin, bone grafts) – climate control as carbon sink – erosion protection of low-lying islands – tourism income • **Threats:** physical damage (anchors, divers, litter, coral extraction), collection of ornaments (CITES Appendix II), sedimentation from rivers and coastal development, pollution (oil, nutrients causing algal shading, sewage, pesticides, sunscreen), ocean acidification, bleaching from warming, overfishing (giant triton / crown of thorns starfish), introduced species (red lionfish), loss of mangroves and seagrass nurseries • **Conservation:** marine protected areas / national parks (Great Barrier Reef, Belize, Chagos), mooring buoys, diver education, No Take Zones which can increase catches elsewhere, sustainable ecotourism and fishing, protection of mangroves and seagrass, erosion and pollution control, planning control of ports and resorts • **Fishing controls:** closed seasons, minimum and maximum catchable sizes, bans on dynamite, trawling and small-mesh nets • **Evaluation:** local measures cannot control global threats (warming, acidification) which need international action – enforcement in remote areas is difficult – conflicts with local livelihoods – large vs small NTZs • Justified conclusion **Also accept**	9

Do not write outside the box

Question	Marking guidance	Marks
	• Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **CITES** → the treaty that controls international trade in endangered species • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen • 'the algae use up all the oxygen' on its own - the oxygen is used by aerobic decomposers breaking down the dead algae	
	Model answer - Level 3 (9 marks)	
	Tropical coral reefs cover about one per cent of the oceans yet support over a quarter of marine species, a diversity built on the symbiosis between polyps and zooxanthellae that recycles nutrients in nutrient-poor water. They matter to people for fisheries of fish, lobsters and conch; for medicines such as AZT, Ara-C, dolastatin, secosteroids, bryostatin and trabectedin discovered in reef organisms; as a carbon sink, since every carbonate unit in coral skeleton absorbs a molecule of carbon dioxide; for dissipating wave energy that would erode low-lying islands; and for tourism income.	
	They are also unusually vulnerable because they need narrow abiotic conditions. Anchors, divers, litter and coral extraction kill polyps; ornaments are collected; sediment from rivers and coastal development smothers them; oil, sewage and nutrients cause algal shading; ocean acidification hinders skeleton formation; warming causes bleaching; overfishing of giant tritons lets crown-of-thorns starfish destroy reefs; introduced lionfish eat native fish; and loss of mangroves and seagrass removes the nurseries and sediment traps reefs depend on.	
	Marine protected areas such as the Great Barrier Reef, Belize and Chagos combine mooring buoys, diver education, no-take zones that raise catches outside them, sustainable tourism and protection of linked habitats, while fishing controls set closed seasons, minimum and maximum sizes and ban dynamite, trawling and small mesh. These work well against local threats where they are enforced, and no-take zones have measurable benefits, but they cannot touch warming and acidification, which need international action, enforcement in remote areas is weak, and restrictions conflict with local livelihoods. Local conservation buys time; only global climate action can secure the reefs.	
	Weaker answer - Level 1 (3 marks)	
	Coral reefs are important because lots of fish live there and people catch them for food. Tourists visit reefs to dive and this brings money. Reefs also protect the coast from waves. Reefs are threatened by climate change which causes bleaching, and by pollution and damage from boats. They are conserved by making marine protected areas where fishing is banned. This helps but climate change is still a problem.	
	What makes the difference	
	• Level 1 has the right headings but Level 3 explains each (why reefs are diverse, why they are a carbon sink, how each threat acts).	
	• Level 3 uses named reserves and specific management measures; Level 1 gives one generic measure.	

Do not write outside the box

Question	Marking guidance	Marks
	• Level 3's evaluation separates local from global threats and reaches a nuanced judgement; Level 1 states that climate change is 'still a problem'.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	
11.1	**Discuss the extent to which the principles of the circular economy could be used to develop sustainable lifestyles.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Definition of sustainability (Brundtland) – population x per capita impact – dynamic equilibria, feedback, tipping points, diversity • **Principles of natural systems:** renewable low-density energy, low temperatures, material cycles, non-toxic wastes – contrast with linear human systems • **Circular economy principles:** cycling materials, renewable energy, supporting ecosystems, separation of technical and biological materials, diversity, connected systems, design for extended use and re-use, optimum production, new technologies • Application to land use, water, minerals, waste, pollution, energy, food with examples (Kalundborg, Eigg, Ol Pejeta, Rio Bravo, EU initiatives, DC grids) • **Measuring progress:** ecological footprint, biocapacity, global hectares, Earth Overshoot Day, carbon footprint, Living Planet Index • **Barriers:** economics, competition, consumer behaviour, need for international agreement, existing infrastructure • Judgement on the extent achievable	25
	Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion. **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-389; AQA spec 3.6.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Dynamic equilibria and negative feedback in climate, hydrological cycle and populations – positive feedback and tipping points caused by humans • **Energy:** solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process) – renewable energy, heat pumps, bioleaching, fuel cells • **Materials:** cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals – recycling, Cradle to Cradle, industrial symbiosis • **Diversity and stability:** rainforests / reefs vs agroecosystems – diverse energy mixes • Optimum vs maximum production – supporting ecosystems in agriculture, forestry, fisheries, urban design • Case studies and metrics – limits and judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4.	

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